

Assignment 16.4 Ai Assisted Coding

Htno:2303a51286

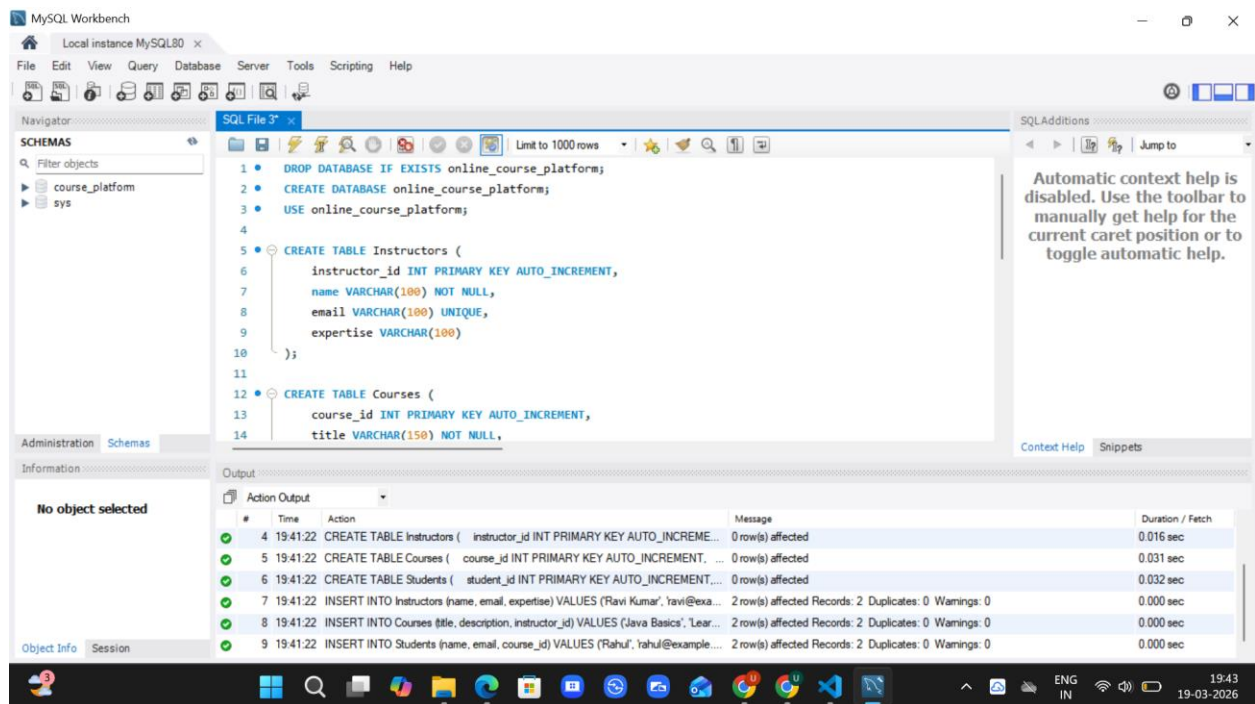
Btno:05

Task 1: Database Schema Design for an Online Course Platform

Prompt:

Generate SQL CREATE TABLE statements for an online course platform including Instructors, Courses, and Students. Ensure the use of primary keys, foreign key relationships, and appropriate data types. The schema should be normalized and logically structured.

Code:



The screenshot shows the MySQL Workbench interface. The main editor window displays the following SQL code:

```
1 DROP DATABASE IF EXISTS online_course_platform;
2 CREATE DATABASE online_course_platform;
3 USE online_course_platform;
4
5 CREATE TABLE Instructors (
6     instructor_id INT PRIMARY KEY AUTO_INCREMENT,
7     name VARCHAR(100) NOT NULL,
8     email VARCHAR(100) UNIQUE,
9     expertise VARCHAR(100)
10 );
11
12 CREATE TABLE Courses (
13     course_id INT PRIMARY KEY AUTO_INCREMENT,
14     title VARCHAR(150) NOT NULL,
```

The left sidebar shows the 'SCHEMAS' panel with 'course_platform' and 'sys' databases listed. The bottom panel shows the 'Output' window with the following results:

#	Time	Action	Message	Duration / Fetch
4	19:41:22	CREATE TABLE Instructors (instructor_id INT PRIMARY KEY AUTO_INCREMENT, ...	0 row(s) affected	0.016 sec
5	19:41:22	CREATE TABLE Courses (course_id INT PRIMARY KEY AUTO_INCREMENT, ...	0 row(s) affected	0.031 sec
6	19:41:22	CREATE TABLE Students (student_id INT PRIMARY KEY AUTO_INCREMENT, ...	0 row(s) affected	0.032 sec
7	19:41:22	INSERT INTO Instructors (name, email, expertise) VALUES ('Ravi Kumar', 'ravi@exa...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
8	19:41:22	INSERT INTO Courses (title, description, instructor_id) VALUES ('Java Basics', 'Lear...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
9	19:41:22	INSERT INTO Students (name, email, course_id) VALUES ('Rahul', 'rahul@example...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec

Output:

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- course_platform
- online_course_platform
- sys

SQL File 3* SQL File 4*

1 • SHOW TABLES;

Result Grid

Tables_in_online_course_platform

- courses
- instructors
- students

Administration Schemas

Information

Schema: online_course_platform

Object Info Session

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5	19:41:22	CREATE TABLE Courses (course_id INT PRIMARY KEY AUTO_INCREMENT, ...	0 row(s) affected	0.031 sec
6	19:41:22	CREATE TABLE Students (student_id INT PRIMARY KEY AUTO_INCREMENT, ...	0 row(s) affected	0.032 sec
7	19:41:22	INSERT INTO Instructors (name, email, expertise) VALUES ('Ravi Kumar', 'ravi@ex...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
8	19:41:22	INSERT INTO Courses (title, description, instructor_id) VALUES ('Java Basics', 'Lea...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
9	19:41:22	INSERT INTO Students (name, email, course_id) VALUES ('Rahul', 'rahul@exampl...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
10	19:46:15	SHOW TABLES	3 row(s) returned	0.000 sec / 0.000 sec

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

ENG IN 19:46 19-03-2026

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- course_platform
- online_course_platform
- sys

SQL File 3* SQL File 4* SQL File 5*

1 • SELECT * FROM Instructors;

Result Grid

instructor_id	name	email	expertise
1	Ravi Kumar	ravi@example.com	Java
2	Anita Sharma	anita@example.com	Python

Administration Schemas

Information

Schema: online_course_platform

Object Info Session

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	19:41:22	CREATE TABLE Students (student_id INT PRIMARY KEY AUTO_INCREMENT, ...	0 row(s) affected	0.032 sec
7	19:41:22	INSERT INTO Instructors (name, email, expertise) VALUES ('Ravi Kumar', 'ravi@ex...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
8	19:41:22	INSERT INTO Courses (title, description, instructor_id) VALUES ('Java Basics', 'Lea...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
9	19:41:22	INSERT INTO Students (name, email, course_id) VALUES ('Rahul', 'rahul@exampl...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
10	19:46:15	SHOW TABLES	3 row(s) returned	0.000 sec / 0.000 sec
11	19:47:16	SELECT * FROM Instructors LIMIT 0, 1000	2 row(s) returned	0.015 sec / 0.000 sec

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

ENG IN 19:47 19-03-2026

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

course_platform

online_course_platform

sys

SQL File 3* SQL File 4* SQL File 5* SQL File 6*

1 • SELECT * FROM Courses;

Result Grid

course_id	title	description	instructor_id
1	Java Basics	Learn Java from scratch	1
2	Python Programming	Learn Python fundamentals	2

Courses 1 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	19:41:22	INSERT INTO Instructors (name, email, expertise) VALUES (Ravi Kumar, 'ravi@ex...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
8	19:41:22	INSERT INTO Courses (title, description, instructor_id) VALUES (Java Basics', Lea...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
9	19:41:22	INSERT INTO Students (name, email, course_id) VALUES (Rahul', 'rahul@exampl...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
10	19:46:15	SHOW TABLES	3 row(s) returned	0.000 sec / 0.000 sec
11	19:47:16	SELECT * FROM Instructors LIMIT 0, 1000	2 row(s) returned	0.015 sec / 0.000 sec
12	19:47:48	SELECT * FROM Courses LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

ENG IN 19:47 19-03-2026

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

course_platform

online_course_platform

sys

SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7*

1 • SELECT * FROM Students;

Result Grid

student_id	name	email	course_id
1	Rahul	rahul@example.com	1
2	Sneha	sneha@example.com	2

Students 1 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	19:41:22	INSERT INTO Courses (title, description, instructor_id) VALUES (Java Basics', Lea...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
9	19:41:22	INSERT INTO Students (name, email, course_id) VALUES (Rahul', 'rahul@exampl...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
10	19:46:15	SHOW TABLES	3 row(s) returned	0.000 sec / 0.000 sec
11	19:47:16	SELECT * FROM Instructors LIMIT 0, 1000	2 row(s) returned	0.015 sec / 0.000 sec
12	19:47:48	SELECT * FROM Courses LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
13	19:48:11	SELECT * FROM Students LIMIT 0, 1000	2 row(s) returned	0.016 sec / 0.000 sec

Object Info Session

ENG IN 19:48 19-03-2026

Explanation:

-->Database Design

Created tables for Instructors, Courses, Students using CREATE TABLE with proper data types and primary keys.

-->Relationships (Foreign Keys)

Connected tables using foreign keys (e.g., course linked to instructor) to maintain data integrity.

-->Normalization

Structured tables without duplication (separate tables for each entity) to follow basic normalization rules.

Task 2: Populate the Database with Sample Records

Prompt:

Generate SQL INSERT statements to populate the database with sample records including at least **3 instructors, 4 courses, and 3 students**, and verify the inserted data using SELECT queries.

Code:

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- course_platform
- online_course_platform
- sys

Administration Schemas

Information

Schema: online_course_platform

Object Info Session

SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9*

Limit to 1000 rows

```

1 -- Step 1: Drop database if already exists
2 DROP DATABASE IF EXISTS online_course;
3
4 -- Step 2: Create fresh database
5 CREATE DATABASE online_course;
6
7 -- Step 3: Use database
8 USE online_course;
9
10 -- Step 4: Create Tables (needed for Task 2)
11 CREATE TABLE Instructors (
12     instructor_id INT PRIMARY KEY,
13     name VARCHAR(50),
14     email VARCHAR(50),

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
34	20:08:33	INSERT INTO Courses VALUES (101, 'Java Programming', 1), (102, 'Python Basics...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.015 sec
35	20:08:33	INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'pri...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
36	20:08:33	show tables	3 row(s) returned	0.000 sec / 0.000 sec
37	20:08:33	SELECT * FROM Instructors LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
38	20:08:33	SELECT * FROM Courses LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
39	20:08:33	SELECT * FROM Students LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

20:10 19-03-2026

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- course_platform
- online_course_platform
- sys

Administration Schemas

Information

Schema: online_course_platform

Object Info Session

SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9*

Limit to 1000 rows

```

39 (102, 'Python Basics', 2),
40 (103, 'Database Systems', 3),
41 (104, 'Web Development', 1);
42
43 INSERT INTO Students VALUES
44 (201, 'Rahul', 'rahul@gmail.com'),
45 (202, 'Priya', 'priya@gmail.com'),
46 (203, 'Arjun', 'arjun@gmail.com');
47
48 -- Step 6: Verify
49 SELECT * FROM Instructors;
50 SELECT * FROM Courses;
51 SELECT * FROM Students;

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
34	20:08:33	INSERT INTO Courses VALUES (101, 'Java Programming', 1), (102, 'Python Basics...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.015 sec
35	20:08:33	INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'pri...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
36	20:08:33	show tables	3 row(s) returned	0.000 sec / 0.000 sec
37	20:08:33	SELECT * FROM Instructors LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
38	20:08:33	SELECT * FROM Courses LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
39	20:08:33	SELECT * FROM Students LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

20:10 19-03-2026

Output:

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- course_platform
- online_course_platform
- sys

SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* x SQL File 9*

Limit to 1000 rows

```
2 DROP DATABASE IF EXISTS online_course;
3
4 -- Step 2: Create fresh database
5 CREATE DATABASE online_course;
6
7 -- Step 3: Use database
```

Result Grid

	instructor_id	name	email	country
1	1	Ravi Kumar	ravi@gmail.com	India
2	2	John Smith	john@gmail.com	USA
3	3	Anita Sharma	anita@gmail.com	India

Administration Schemas

Information

Schema: online_course_platform

Object Info Session

Result 4 Instructors 5 Courses 6 Students 7

Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
34	20:08:33	INSERT INTO Courses VALUES (101, 'Java Programming', 1), (102, 'Python Basics...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.015 sec
35	20:08:33	INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'pri...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
36	20:08:33	show tables	3 row(s) returned	0.000 sec / 0.000 sec
37	20:08:33	SELECT * FROM Instructors LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
38	20:08:33	SELECT * FROM Courses LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
39	20:08:33	SELECT * FROM Students LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- course_platform
- online_course_platform
- sys

SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* x SQL File 9*

Limit to 1000 rows

```
2 DROP DATABASE IF EXISTS online_course;
3
4 -- Step 2: Create fresh database
5 CREATE DATABASE online_course;
6
7 -- Step 3: Use database
```

Result Grid

	course_id	course_name	instructor_id
1	101	Java Programming	1
2	102	Python Basics	2
3	103	Database Systems	3
4	104	Web Development	1

Administration Schemas

Information

Schema: online_course_platform

Object Info Session

Result 4 Instructors 5 Courses 6 Students 7

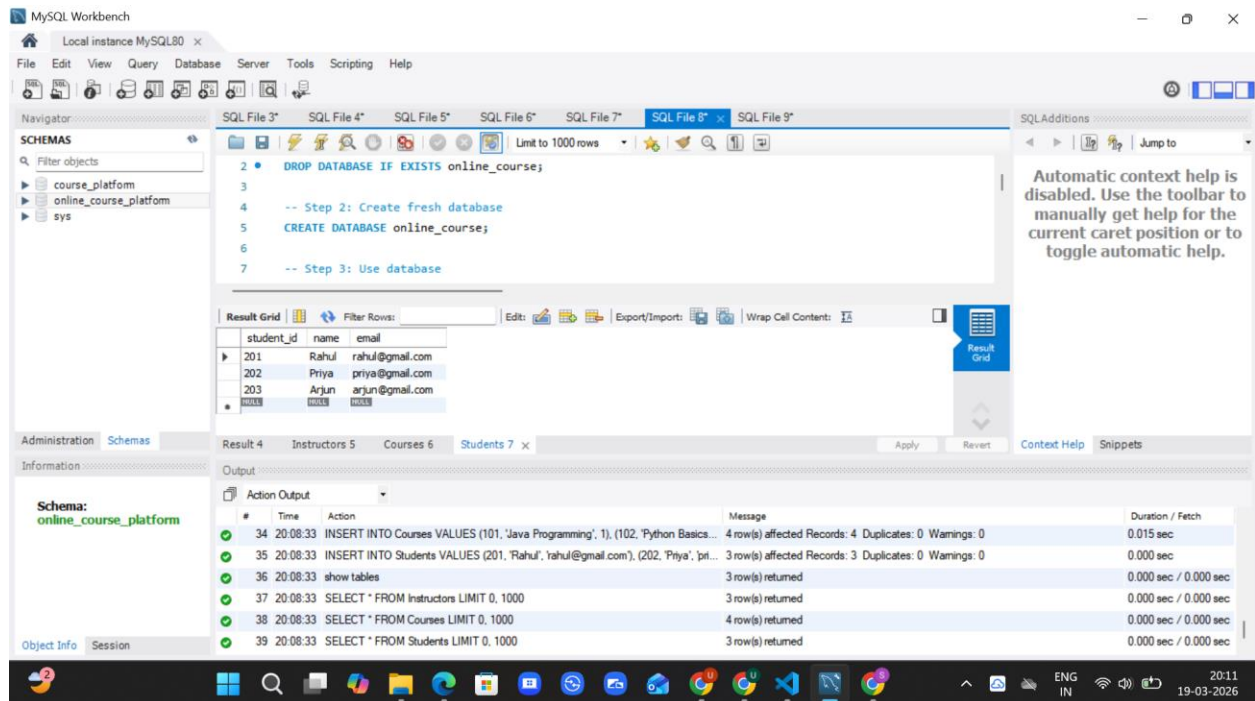
Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
34	20:08:33	INSERT INTO Courses VALUES (101, 'Java Programming', 1), (102, 'Python Basics...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.015 sec
35	20:08:33	INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'pri...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
36	20:08:33	show tables	3 row(s) returned	0.000 sec / 0.000 sec
37	20:08:33	SELECT * FROM Instructors LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
38	20:08:33	SELECT * FROM Courses LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
39	20:08:33	SELECT * FROM Students LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.



Explanation:

-->Used DROP DATABASE to remove old data and start fresh

-->Recreated tables and inserted sample records

-->Verified data using SELECT queries

Task 3: Implement Core Database Operations (CRUD)

Prompt:

Perform SQL operations to **retrieve all courses, update a student's email, and delete a course**, then verify using SELECT queries.

Code:

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- course_platform
- online_course_platform
- sys

Administration Schemas

Information

Schema: online_course_platform

Object Info Session

SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11*

Limit to 1000 rows

```
1 -- Step 1: Drop database to avoid already exists error
2 DROP DATABASE IF EXISTS online_course;
3
4 -- Step 2: Create database
5 CREATE DATABASE online_course;
6
7 -- Step 3: Use database
8 USE online_course;
9
10 -- Step 4: Create tables
11 CREATE TABLE Instructors (
12     instructor_id INT PRIMARY KEY,
13     name VARCHAR(50),
14     email VARCHAR(50),
```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
48	20:13:42	INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'priya@gmail.com');	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
49	20:13:42	SELECT * FROM Courses LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
50	20:13:42	UPDATE Students SET email = 'rahul_new@gmail.com' WHERE student_id = 201	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
51	20:13:42	DELETE FROM Courses WHERE course_id = 104	1 row(s) affected	0.000 sec
52	20:13:42	SELECT * FROM Students LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
53	20:13:42	SELECT * FROM Courses LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

20:14 19-03-2026

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- course_platform
- online_course_platform
- sys

Administration Schemas

Information

Schema: online_course_platform

Object Info Session

SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11*

Limit to 1000 rows

```
31 -- Step 5: Insert sample data
32 INSERT INTO Instructors VALUES
33     (1, 'Ravi Kumar', 'ravi@gmail.com', 'India'),
34     (2, 'John Smith', 'john@gmail.com', 'USA'),
35     (3, 'Anita Sharma', 'anita@gmail.com', 'India');
36
37 INSERT INTO Courses VALUES
38     (101, 'Java Programming', 1),
39     (102, 'Python Basics', 2),
40     (103, 'Database Systems', 3),
41     (104, 'Web Development', 1);
42
43 INSERT INTO Students VALUES
44     (201, 'Rahul', 'rahul@gmail.com'),
```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
48	20:13:42	INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'priya@gmail.com');	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
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20:14 19-03-2026

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

Schemas

- course_platform
- online_course_platform
- sys

SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11*

Limit to 1000 rows

```

52
53 -- Update student email
54 • UPDATE Students
55   SET email = 'rahul_new@gmail.com'
56   WHERE student_id = 201;
57
58 -- Delete a course
59 • DELETE FROM Courses
60   WHERE course_id = 104;
61
62 -- Verify changes
63 • SELECT * FROM Students;
64 • SELECT * FROM Courses;

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Administration Schemas

Information

Schema: online_course_platform

Output

Action Output

#	Time	Action	Message	Duration / Fetch
48	20:13:42	INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'priya@gmail.com');	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
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51	20:13:42	DELETE FROM Courses WHERE course_id = 104	1 row(s) affected	0.000 sec
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53	20:13:42	SELECT * FROM Courses LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

ENG IN 20:14 19-03-2026

Output:

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- course_platform
- online_course_platform
- sys

SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* x SQL File 11*

59 • DELETE FROM Courses
60 WHERE course_id = 104;
61
62 -- Verify changes
63 • SELECT * FROM Students;
64 • SELECT * FROM Courses;

Result Grid

course_id	course_name	instructor_id
101	Java Programming	1
102	Python Basics	2
103	Database Systems	3
104	Web Development	1

Administration Schemas

Information

Schema: online_course_platform

Object Info Session

Output

Action Output

#	Time	Action	Message	Duration / Fetch
48	20:13:42	INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'priya@gmail.com'), (203, 'Arjun', 'arjun@gmail.com');	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
49	20:13:42	SELECT * FROM Courses LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
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ENG IN 20:15 19-03-2026

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- course_platform
- online_course_platform
- sys

SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* x SQL File 11*

59 • DELETE FROM Courses
60 WHERE course_id = 104;
61
62 -- Verify changes
63 • SELECT * FROM Students;
64 • SELECT * FROM Courses;

Result Grid

student_id	name	email
201	Rahul	rahul_new@gmail.com
202	Priya	priya@gmail.com
203	Arjun	arjun@gmail.com

Administration Schemas

Information

Schema: online_course_platform

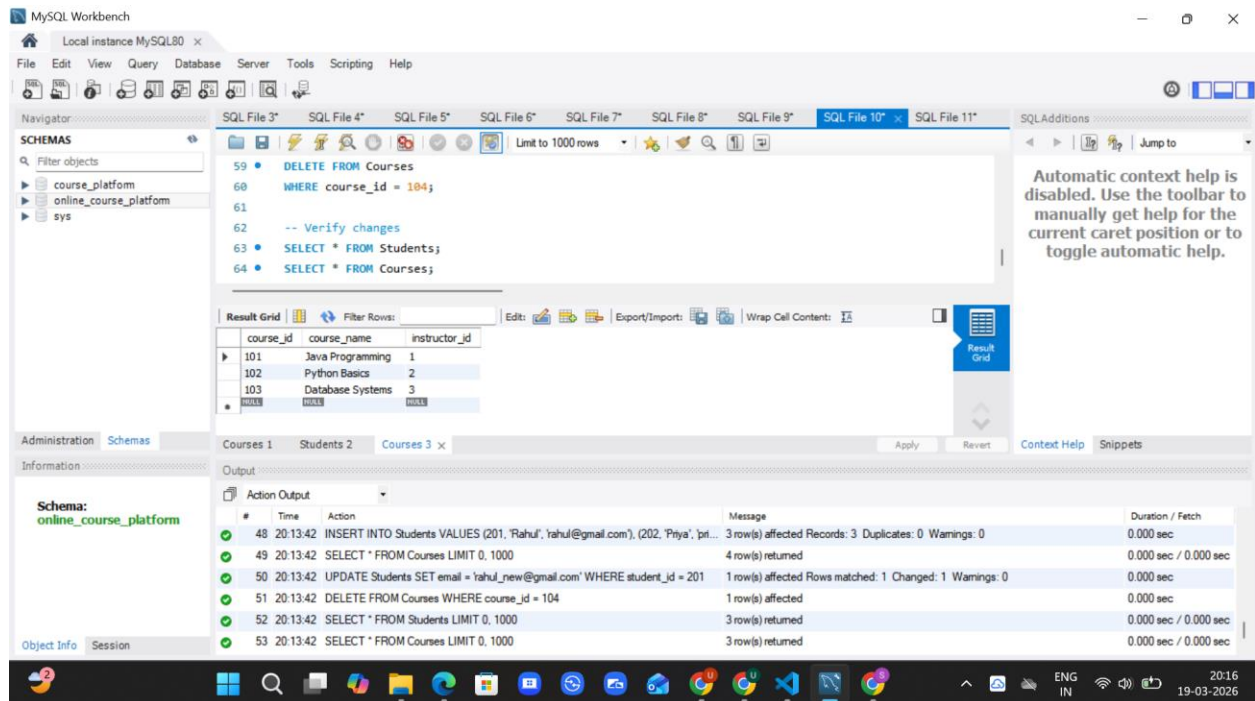
Object Info Session

Output

Action Output

#	Time	Action	Message	Duration / Fetch
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49	20:13:42	SELECT * FROM Courses LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
50	20:13:42	UPDATE Students SET email = 'rahul_new@gmail.com' WHERE student_id = 201	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
51	20:13:42	DELETE FROM Courses WHERE course_id = 104	1 row(s) affected	0.000 sec
52	20:13:42	SELECT * FROM Students LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
53	20:13:42	SELECT * FROM Courses LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

ENG IN 20:15 19-03-2026



Explanation:

-->Used SELECT to retrieve all course records

-->Updated student email using UPDATE statement

-->Deleted a course using DELETE and verified changes

Task 4: Reporting Using Aggregate Queries

Prompt:

Generate SQL queries to **count students per course** and **list courses with more than a specified number of enrollments** using GROUP BY and HAVING.

Code:

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

course_platform

online_course_platform

sys

Administration Schemas

Information

Schema: online_course_platform

Object Info Session

SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*

Limit to 1000 rows

1 -- Step 1: Drop database (clean start)

2 DROP DATABASE IF EXISTS online_course;

3

4 -- Step 2: Create database

5 CREATE DATABASE online_course;

6

7 -- Step 3: Use database

8 USE online_course;

9

10 -- Step 4: Create tables

11 CREATE TABLE Instructors (

12 instructor_id INT PRIMARY KEY,

13 name VARCHAR(50),

14 email VARCHAR(50),

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
61	20:32:53	INSERT INTO Instructors VALUES (1, 'Ravi Kumar', 'ravi@gmail.com', 'India'), (2, 'J...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
62	20:32:53	INSERT INTO Courses VALUES (101, 'Java Programming', 1), (102, 'Python Basics...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
63	20:32:53	INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'pri...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
64	20:32:53	INSERT INTO Enrollments VALUES (1, 201, 101), (2, 202, 101), (3, 203, 102), (4, 2...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
65	20:32:53	SELECT course_id, COUNT(student_id) AS total_students FROM Enrollments GRO...	3 row(s) returned	0.000 sec / 0.000 sec
66	20:32:53	SELECT course_id, COUNT(student_id) AS total_students FROM Enrollments GRO...	2 row(s) returned	0.000 sec / 0.000 sec

26°

ENG IN

20:34

19-03-2026

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

course_platform

online_course_platform

sys

Administration Schemas

Information

Schema: online_course_platform

Object Info Session

SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13*

Limit to 1000 rows

64

65 -- Step 6: Aggregate Queries

66

67 -- Count students per course

68 SELECT course_id, COUNT(student_id) AS total_students

69 FROM Enrollments

70 GROUP BY course_id;

71

72 -- Courses with more than 1 student

73 SELECT course_id, COUNT(student_id) AS total_students

74 FROM Enrollments

75 GROUP BY course_id

76 HAVING COUNT(student_id) > 1;

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
61	20:32:53	INSERT INTO Instructors VALUES (1, 'Ravi Kumar', 'ravi@gmail.com', 'India'), (2, 'J...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
62	20:32:53	INSERT INTO Courses VALUES (101, 'Java Programming', 1), (102, 'Python Basics...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
63	20:32:53	INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'pri...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
64	20:32:53	INSERT INTO Enrollments VALUES (1, 201, 101), (2, 202, 101), (3, 203, 102), (4, 2...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
65	20:32:53	SELECT course_id, COUNT(student_id) AS total_students FROM Enrollments GRO...	3 row(s) returned	0.000 sec / 0.000 sec
66	20:32:53	SELECT course_id, COUNT(student_id) AS total_students FROM Enrollments GRO...	2 row(s) returned	0.000 sec / 0.000 sec

ENG IN

20:33

19-03-2026

Output:

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

course_platform
online_course_platform
sys

SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* x SQL File 13*

Limit to 1000 rows

SQL File 12* x

SQL File 13*

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

71
72 -- Courses with more than 1 student
73 • SELECT course_id, COUNT(student_id) AS total_students
74 FROM Enrollments
75 GROUP BY course_id
76 HAVING COUNT(student_id) > 1;

```

Result Grid

course_id	total_students
101	2
102	1
103	2

Administration Schemas

Information

Schema: online_course_platform

Object Info Session

Output

Action Output

#	Time	Action	Message	Duration / Fetch
61	20:32:53	INSERT INTO Instructors VALUES (1, 'Ravi Kumar', 'ravi@gmail.com', 'India'), (2, 'J...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
62	20:32:53	INSERT INTO Courses VALUES (101, 'Java Programming', 1), (102, 'Python Basics...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
63	20:32:53	INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'pri...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
64	20:32:53	INSERT INTO Enrollments VALUES (1, 201, 101), (2, 202, 101), (3, 203, 102), (4, 2...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
65	20:32:53	SELECT course_id, COUNT(student_id) AS total_students FROM Enrollments GRO...	3 row(s) returned	0.000 sec / 0.000 sec
66	20:32:53	SELECT course_id, COUNT(student_id) AS total_students FROM Enrollments GRO...	2 row(s) returned	0.000 sec / 0.000 sec

26° ENG IN 20:35 19-03-2026

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

course_platform
online_course_platform
sys

SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* x SQL File 13*

Limit to 1000 rows

SQL File 12* x

SQL File 13*

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

71
72 -- Courses with more than 1 student
73 • SELECT course_id, COUNT(student_id) AS total_students
74 FROM Enrollments
75 GROUP BY course_id
76 HAVING COUNT(student_id) > 1;

```

Result Grid

course_id	total_students
101	2
103	2

Administration Schemas

Information

Schema: online_course_platform

Object Info Session

Output

Action Output

#	Time	Action	Message	Duration / Fetch
61	20:32:53	INSERT INTO Instructors VALUES (1, 'Ravi Kumar', 'ravi@gmail.com', 'India'), (2, 'J...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
62	20:32:53	INSERT INTO Courses VALUES (101, 'Java Programming', 1), (102, 'Python Basics...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
63	20:32:53	INSERT INTO Students VALUES (201, 'Rahul', 'rahul@gmail.com'), (202, 'Priya', 'pri...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
64	20:32:53	INSERT INTO Enrollments VALUES (1, 201, 101), (2, 202, 101), (3, 203, 102), (4, 2...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
65	20:32:53	SELECT course_id, COUNT(student_id) AS total_students FROM Enrollments GRO...	3 row(s) returned	0.000 sec / 0.000 sec
66	20:32:53	SELECT course_id, COUNT(student_id) AS total_students FROM Enrollments GRO...	2 row(s) returned	0.000 sec / 0.000 sec

26° ENG IN 20:35 19-03-2026

Explanation:

-->Used GROUP BY to count students in each course

-->Used COUNT() function for aggregation

-->Used HAVING to filter courses with more than 1 student

Task 5: Query Optimization with AI Suggestions

Prompt:

Analyze and optimize a query that retrieves courses taught by instructors from a specific country using **efficient joins and indexing**, and compare results.

Code:

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

- course_platform
- online_course_platform
- sys

SCHEMAS

Administration Schemas

Information

Schema: online_course_platform

Object Info Session

SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13* x SQL File 14*

Limit to 1000 rows

```

1 -- Step 1: Drop database (to avoid already exists)
2 DROP DATABASE IF EXISTS online_course;
3
4 -- Step 2: Create database
5 CREATE DATABASE online_course;
6
7 -- Step 3: Use database
8 USE online_course;
9
10 -- Step 4: Create tables
11 CREATE TABLE Instructors (
12     instructor_id INT PRIMARY KEY,
13     name VARCHAR(50),
14     email VARCHAR(50),

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
72	20:41:57	INSERT INTO Instructors VALUES (1, 'Ravi Kumar', 'ravi@gmail.com', 'India'), (2, 'J...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
73	20:41:57	INSERT INTO Courses VALUES (101, 'Java Programming', 1), (102, 'Python Basics...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
74	20:41:57	SELECT course_name FROM Courses WHERE instructor_id IN (SELECT instru...	3 row(s) returned	0.000 sec / 0.000 sec
75	20:41:57	SELECT c.course_name FROM Courses c JOIN Instructors i ON c.instructor_id = i.i...	3 row(s) returned	0.000 sec / 0.000 sec
76	20:41:57	CREATE INDEX idx_country ON Instructors(country)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
77	20:41:57	CREATE INDEX idx_instructor ON Courses(instructor_id)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec

20:43 19-03-2026

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

- course_platform
- online_course_platform
- sys

SCHEMAS

Administration Schemas

Information

Schema: online_course_platform

Object Info Session

SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL File 13* x SQL File 14*

Limit to 1000 rows

```

43 WHERE country = 'India'
44 );
45
46 -- Step 7: Optimized Query (Using JOIN)
47 SELECT c.course_name
48 FROM Courses c
49 JOIN Instructors i
50 ON c.instructor_id = i.instructor_id
51 WHERE i.country = 'India';
52
53 -- Step 8: Add Index (Optimization)
54 CREATE INDEX idx_country ON Instructors(country);
55 CREATE INDEX idx_instructor ON Courses(instructor_id);

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

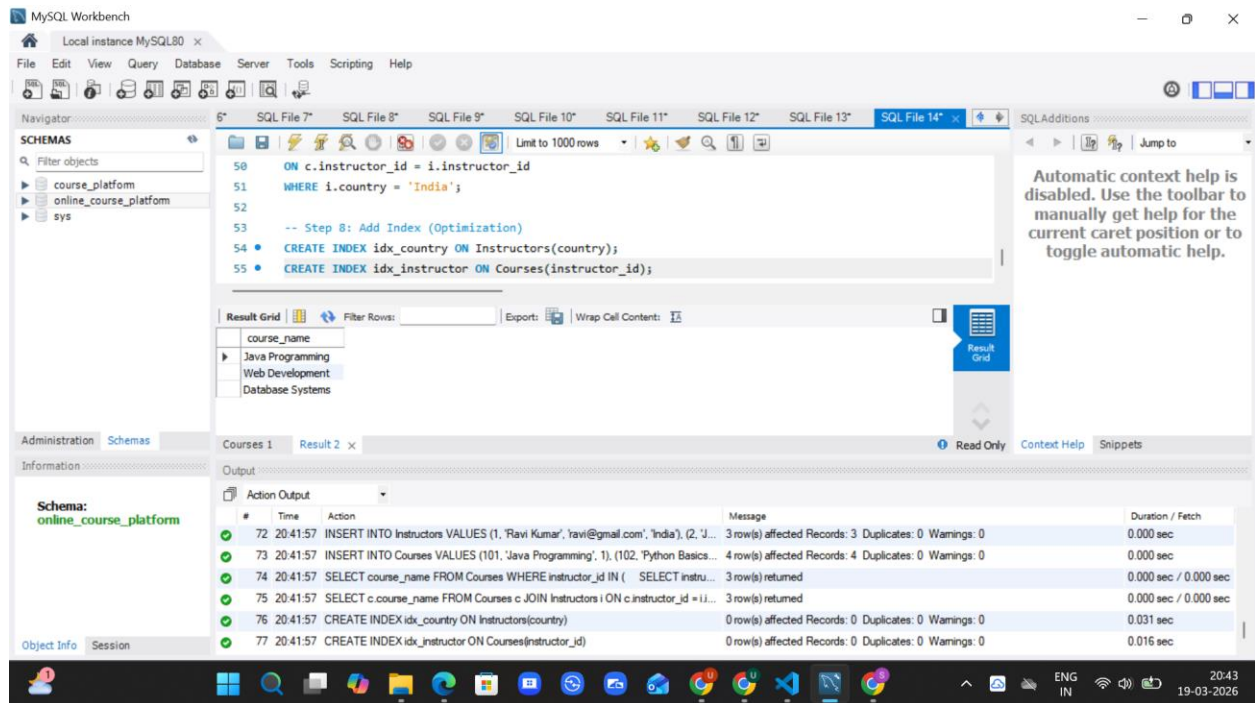
Output

Action Output

#	Time	Action	Message	Duration / Fetch
72	20:41:57	INSERT INTO Instructors VALUES (1, 'Ravi Kumar', 'ravi@gmail.com', 'India'), (2, 'J...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
73	20:41:57	INSERT INTO Courses VALUES (101, 'Java Programming', 1), (102, 'Python Basics...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
74	20:41:57	SELECT course_name FROM Courses WHERE instructor_id IN (SELECT instru...	3 row(s) returned	0.000 sec / 0.000 sec
75	20:41:57	SELECT c.course_name FROM Courses c JOIN Instructors i ON c.instructor_id = i.i...	3 row(s) returned	0.000 sec / 0.000 sec
76	20:41:57	CREATE INDEX idx_country ON Instructors(country)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
77	20:41:57	CREATE INDEX idx_instructor ON Courses(instructor_id)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec

20:43 19-03-2026

Output:



Explanation:

-->Used a normal query with subquery to fetch courses by instructor country

-->Optimized using JOIN for better performance and readability

-->Added indexing to improve query speed on large datasets